

IIC control servo

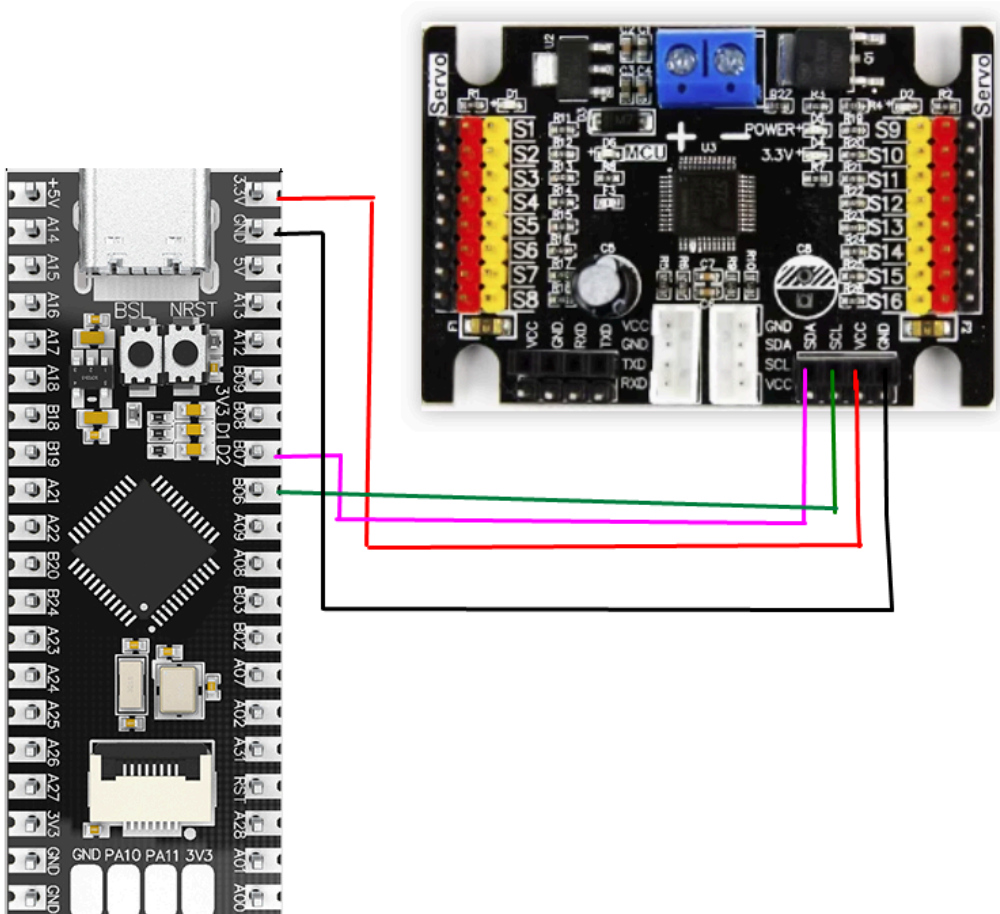
1. Learning objectives

Control the servo through the IIC of the 16-channel servo driver board

2. Hardware connection

The servo used in this case is a 180-degree servo, which is connected to the S1 pin of the 16-channel servo driver board

Connecting MSPM0G3519 with the 16-channel servo driver board



All modules should be connected together with a common ground line

3. Main program description

- bsp_i2c_gpio.h

```
#define I2C_ADDR                0x5A

#define macI2C_WR    0        /* 写控制bit    write control bit */
#define macI2C_RD    1        /* 读控制bit    Read control bit */
```

```

/* 定义I2C总线连接的GPIO端口，用户只需要修改下面4行代码即可任意改变SCL和SDA的引脚 */
/* Define the GPIO port connected to the I2C bus. Users only need to modify the following 4
lines of code to change the SCL and SDA pins at will */
#define macGPIO_PORT_I2C      GPIOB          /* GPIO端口 GPIO Ports*/
#define macRCC_I2C_PORT      RCC_APB2Periph_GPIOB      /* GPIO端口时钟 GPIO port clock*/
#define macI2C_SCL_PIN      SCL_PIN_6_PIN      /* 连接到SCL时钟线的GPIO GPIO connected to
SCL clock line*/
#define macI2C_SDA_PIN      SDA_PIN_7_PIN      /* 连接到SDA数据线的GPIO GPIO connected to
SDA data line*/

#define macI2C_SCL_1()  DL_GPIO_setPins(macGPIO_PORT_I2C, macI2C_SCL_PIN)      /* SCL = 1
*/
#define macI2C_SCL_0()  DL_GPIO_clearPins(macGPIO_PORT_I2C, macI2C_SCL_PIN)    /* SCL = 0
*/

#define macI2C_SDA_1()  DL_GPIO_setPins(macGPIO_PORT_I2C, macI2C_SDA_PIN)      /* SDA = 1
*/
#define macI2C_SDA_0()  DL_GPIO_clearPins(macGPIO_PORT_I2C, macI2C_SDA_PIN)    /* SDA = 0
*/

#define macI2C_SDA_READ()  DL_GPIO_readPins(macGPIO_PORT_I2C, macI2C_SDA_PIN)  /* 读SDA口线
状态 Read SDA line status*/
#define macI2C_SCL_READ()  DL_GPIO_readPins(macGPIO_PORT_I2C, macI2C_SCL_PIN)  /* 读SCL口线
状态 Read SCL line status*/

void Start_I2c(void);
void Stop_I2c(void);
void I2C_SendByte(unsigned char c);
unsigned char I2C_RcvByte(void);
void Ack_I2c(uint8_t a);

uint8_t IIC_Servo(unsigned char servonum,unsigned char angle);

void delay(int z);

#endif

```

Defines the I2C port and its operation function, and declares the basic I2C operation function and servo control function.

- bsp_i2c_gpio.c

```

uint8_t IIC_Servo(unsigned char servonum,unsigned char angle)
{
    Start_I2c();          //启动总线  Start bus
    I2C_SendByte(I2C_ADDR); //发送器件地址  Send device address
    if(ack==0)return(0);
}

```

```

    I2C_SendByte(servonum);    //发送数据  Send data
    if(ack==0)return(0);
    I2C_SendByte(angle);      //发送数据  Send data
    if(ack==0)return(0);

    Stop_I2c();                //结束总线  End the bus
    delay_ms(100);
    return(1);
}

```

Define i2c control servo function. Servo range (1-16), angle (0-180).

- empty.c

```

int main(void)
{
    int i=0;
    SYSCFG_DL_init();

    while (1)
    {
        for(i = 0;i<180;i+=5){
            IIC_Servo(1,i);
        }
        IIC_Servo(1,0);
    }
}

```

Control the No. 1 servo to rotate from 0 to 180 degrees, and then return to 0 degrees after reaching 180 degrees, and repeat this process.

Note: The project source code must be placed in the SDK path for compilation,

For example, the path: D:\TI\M0_SDK\mspm0_sdk_1_30_00_03\1.TB6612

新加卷 (D:) > TI > M0_SDK > mspm0_sdk_1_30_00_03				
名称	修改日期	类型	大小	
1.TB6612	2024/7/22 18:59	文件夹		
2.AT8236	2024/7/22 19:47	文件夹		
3.Enconder	2024/7/23 10:36	文件夹		
4.Servo	2024/7/23 11:13	文件夹		
docs	2024/7/23 10:33	文件夹		
examples	2024/7/23 10:34	文件夹		
kernel	2024/7/23 10:37	文件夹		
source	2024/7/23 10:33	文件夹		
tools	2024/7/23 10:33	文件夹		
imports.mak	2024/1/25 11:45	MAK 文件	2 KB	
known_issues_FAQ.html	2024/1/25 11:42	Microsoft Edge ...	67 KB	
license_mspm0_sdk_1_30_00_03.txt	2024/1/25 11:42	文本文档	33 KB	
manifest_mspm0_sdk_1_30_00_03.html	2024/1/25 11:42	Microsoft Edge ...	113 KB	
mspm0sdk_1_30_00_03.log	2024/7/23 10:42	文本文档	5,237 KB	
release_notes_mspm0_sdk_1_30_00_0...	2024/1/25 11:42	Microsoft Edge ...	108 KB	
uninstall.dat	2024/7/23 10:39	DAT 文件	344 KB	
uninstall.exe	2024/7/23 10:39	应用程序	6,048 KB	

4. Experimental phenomenon

Connect the servo to the S1 interface of the 16-channel servo board, burn the iic control program to MSPM0G3519, and connect the 16-channel servo driver board and MSPM0G3519 according to the wiring diagram. After power on, you can see that the No. 1 servo rotates from 0 degrees to 180 degrees, and then returns to 0 degrees, and repeats this process.